

Team Notebook

rag_korlacodes2pdf. — —authorShahjalal

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1 build

```
{
"cmd" : ["g++ -std=c++20 $file_name -o $file_base_name &&
        timeout 5s ./ $file_base_name<input.txt>output.txt"],
"selector" : "source.cpp",
"shell": true,
//"working_dir" : "$file_path"
}
```

2 DP

2.1 sos

```
// sum over subsets
for (int i = 0; i < B; i++) {
    for (int mask = 0; mask < (1 << B); mask++) {
        if ((mask & (1 << i)) != 0) {
            f[mask] += f[mask ^ (1 << i)];
        }
    }
}

// sum over supersets
for (int i = 0; i < B; i++) {
    for (int mask = (1 << B) - 1; mask >= 0; mask--) {
        if ((mask & (1 << i)) == 0) g[mask] += g[mask ^ (1 << i)];
    }
}
```

3 DS

3.1 BIT

```
struct BIT {
    ll n;
    vector<ll> bit;
    BIT(ll _n) : n(_n), bit(n + 1) {}
    void add(ll x, ll val) {
        for (ll i = x; i <= n; i += (i & -i)) bit[i] += val;
    }
    ll sum(ll x) {
        ll ans = 0;

```

```
        for (ll i = x; i > 0; i -= (i & -i)) ans += bit[i];
        return ans;
    }
    ll query(ll l, ll r) {
        return sum(r) - sum(l - 1);
    }
};

struct BIT2D {
    ll n, m;
    vector<vector<ll>> bit;
    BIT2D(ll _n, ll _m) : n(_n), m(_m), bit(n + 1, vector<ll>
        >(m + 1)) {}
    void add(ll x, ll y, ll val) {
        for (ll i = x; i <= n; i += (i & -i)) {
            for (ll j = y; j <= m; j += (j & -j)) {
                bit[i][j] += val;
            }
        }
    }
    ll sum(ll x, ll y) {
        ll ans = 0;
        for (ll i = x; i > 0; i -= (i & -i)) {
            for (ll j = y; j > 0; j -= (j & -j)) {
                ans += bit[i][j];
            }
        }
        return ans;
    }
    ll query(ll x1, ll y1, ll x2, ll y2) {
        return sum(x2, y2) - sum(x1 - 1, y2) - sum(x2, y1 - 1)
            + sum(x1 - 1, y1 - 1);
    }
};
```

3.2 Centroid Decom

```
typedef struct node {
    vector<int> adj;
    int sub=1;int p;int pp;int dep=0;int vis=0;
} node;
node tree[100010];
vector<vector<pair<int,ll>>> grp(100010);
int gp=0;
int root;
int n; ll m;
ll ans = 0;
void dfs(int u,int p) {
    tree[u].sub = 1;
    for (auto v : tree[u].adj) {

```

```
        if(tree[v].vis != 1 && v != p) {
            dfs(v,u);
            tree[u].sub += tree[v].sub;
        }
    }
}

void nextlev(int u, int p, int r, int d) {
    tree[u].pp = p;
    tree[u].dep = d;
    ll eitar = 0;
    if(p==r) {gp++; grp[gp].clear();}
    if(p != -1) {
        grp[gp].push_back({u,edge});
    }
    for(int i = 0 ; i < tree[u].adj.size();i++) {
        int v = tree[u].adj[i];
        if(v!=p && tree[v].vis != 1) nextlev(v,u,r,d+1);
    }
}

void solve(int c) {
    map<ll,int> mymap;
    val[c] = 0;
    f(i,gp) for(auto& v : grp[i+1]) mymap[ultaval[v.first]]++;
    ans += (ll)mymap[0];
    f(i,gp) {
        for(auto& v : grp[i+1]) mymap[ultaval[v.first]]--;
        // ll eita = 0;
        for(int j = 0; j < grp[i+1].size(); j++) {
            }
        for(auto& v : grp[i+1]) mymap[ultaval[v.first]]++;
    }
}

int dfs2(int u,int p,int n) {
    // printf("u -- %d v-- %d\n",u,v);
    for (auto v : tree[u].adj) {
        if(tree[v].vis != 1 && v != p && tree[v].sub > (n/2))
            { return dfs2(v,u,n);}
    }
    return u;
}

void build(int u, int p) {
    dfs(u,p); //SYBTREE
    int c = dfs2(u,p,tree[u].sub);
    if(p==-1) root=c;
    tree[c].p = p;
    gp=0;

```

```

    nextlev(c,-1,c,0,0,0);
    solve(c);
    tree[c].vis = 1;
    for(auto& v : tree[c].adj) if(tree[v].vis==0) build(v,c);
}

```

3.3 dsu

```

const int N = 2e5 + 5;
vector<int> parent(N), siz(N);
void initi(int n) {
    for (int i = 1; i <= n; ++i) {
        parent[i] = i;
        siz[i] = 1;
    }
}
int findpar(int node) {
    if(node == parent[node]) return node;
    return parent[node] = findpar(parent[node]);
}
void unio(int u, int v) {
    u = findpar(u);
    v = findpar(v);
    if(u == v) return;
    parent[v] = u;
    siz[u] += siz[v];
}

```

3.4 HLD and Merge sort tree

```

typedef struct node {
    int p;
    vector<int> adj;
    int mxchild;
    int chain;
    int ind_in_chain;
    int subtree;
    int koto;
    int price;
} node;
node tree[200010];
vector<int> chain(200010);
vector<int> start_c(200010);
vector<vector<int>> seg(800010);
vector<vector<ll>> sum(800010);
vector<int> dfs_array(200010);
// vector<vector<seg>> segtree(500010);

```

```

int c;
int ind=-1;
void dfs(int u, int p) {
    tree[u].p = p;
    int mx=0;
    f(i, tree[u].adj.size()) {
        int v = tree[u].adj[i];
        if(p==v) continue;
        dfs(v,u);
        tree[u].subtree += tree[v].subtree;
        if(mx < tree[v].subtree) {mx = tree[v].subtree; tree[
            u].mxchild = v;}
    }
}
void hld(int u) {
    dfs_array[++ind] = u;
    chain[c]++;
    // chain[c].push_back(u);
    tree[u].chain = c;
    tree[u].ind_in_chain = chain[c] - 1;
    if(tree[u].mxchild) hld(tree[u].mxchild);
    f(i, tree[u].adj.size()) {
        int v = tree[u].adj[i];
        if(v == tree[u].p || v == tree[u].mxchild) continue;
        c++;
        start_c[c] = ind+1;
        hld(v);
    }
}
void merge(int ind) {
    int lsize = seg[2*ind+1].size();
    int rsize = seg[2*ind+2].size();
    seg[ind].clear();
    sum[ind].clear();
    seg[ind].resize(lsize+rsize);
    sum[ind].resize(lsize+rsize);
    int l = 0, r = 0, i=0;
    while(l < lsize || r < rsize) {
        if(l==lsize) {
            seg[ind][i] =(seg[2*ind+2][r]);r++;
        }
        else if(r==rsize) {
            seg[ind][i] =(seg[2*ind+1][l]);l++;
        }
        else if(seg[2*ind+1][l] < seg[2*ind+2][r]) {
            seg[ind][i] =(seg[2*ind+1][l]);l++;
        }
        else {
            seg[ind][i] =(seg[2*ind+2][r]);r++;
        }
    }
}

```

```

    sum[ind][i] = ((l==0 ? 0 : sum[ind*2+1][l-1]) + (r==0
        ? 0 : sum[ind*2+2][r-1]));
    i++;
}
// return narr;
}
void build(vector<int>& arr, int l, int r, int ind) {
    if(l==r) {
        l = arr[l];
        seg[ind].resize(1);
        sum[ind].resize(1);
        seg[ind][0] = (tree[l].price);
        sum[ind][0] = ((ll)tree[l].koto);
    } else {
        int m = (l+r)/2;
        build(arr,l,m,2*ind+1);
        build(arr,m+1,r,2*ind+2);
        merge(ind);
    }
}
ll query(int s, int e, int l, int r, int ind, int tar) {
    if(s>e) return 0;
    if(s==l && e==r) {
        int ans = -1;
        l = 0, r = seg[ind].size()-1;
        while(l <= r) {
            int m = (l+r)/2;
            if(seg[ind][m] <= tar) {
                ans = m;
                l = m+1;
            } else {
                r = m-1;
            }
        }
        return ans==-1?0:sum[ind][ans];
    } else {
        int m =(l+r)/2;
        return query(s,min(e,m),l,m,2*ind+1,tar) + query(max(
            s,m+1),e,m+1,r,2*ind+2,tar);
    }
}

```

3.5 lazy

```

using ll = long long;
const int N = 1e5 + 5;
const int mod = 1e9 + 7;
vector<ll> t(4 * N, 0), lazy(4 * N, 0);
void push(int ind, int al, int ar) {

```

```

    int mid = al + (ar - al) / 2;
    lazy[2 * ind] += lazy[ind];
    t[2 * ind] += (mid - al + 1) * lazy[ind];
    lazy[2 * ind + 1] += lazy[ind];
    t[2 * ind + 1] += (ar - mid) * lazy[ind];
    lazy[ind] = 0;
}
void update(int ind, int al, int ar, int l, int r, int v) {
    if(r < al or l > ar) return;
    if(l <= al and ar <= r) {
        t[ind] += 1ll * (ar - al + 1) * v;
        lazy[ind] += v;
        return;
    }
    push(ind, al, ar);
    int mid = al + (ar - al) / 2;
    update(2 * ind, al, mid, l, r, v);
    update(2 * ind + 1, mid + 1, ar, l, r, v);
    t[ind] = t[2 * ind] + t[2 * ind + 1];
}
ll query(int ind, int al, int ar, int l, int r) {
    if(r < al or l > ar) return 0;
    if(l <= al and ar <= r) {
        return t[ind];
    }
    push(ind, al, ar);
    int mid = al + (ar - al) / 2;
    ll left = query(2 * ind, al, mid, l, r);
    ll rgt = query(2 * ind + 1, mid + 1, ar, l, r);
    return left + rgt;
}

```

3.6 MOs

```

const int BLOCK_SIZE = 350;
int n, q, ar[N], freq[N], res = 0, ans[10*N];
struct Query {
    int l, r, id;
    bool operator<(const Query &y) const {
        int x_block = l / BLOCK_SIZE;
        int y_block = y.l / BLOCK_SIZE;
        if(x_block == y_block)
            return r < y.r;
        return x_block < y_block;
    }
};
void add(int i) {
    int x = ar[i]; res -= freq[x] / 2; freq[x]++; res += freq[x] / 2;
}

```

```

}
void Remove(int i) {
    int x = ar[i]; res -= freq[x] / 2; freq[x]--; res += freq[x] / 2;
}
void solve() { // need to reset freq if needed
    cin >> n;
    for (int i = 1; i <= n; i++) cin >> ar[i];
    cin >> q;
    vector<Query> qr;
    res = 0; int ii = 0;
    while(q--) {
        int l, r; cin >> l >> r; qr.push_back(Query({l, r, ii++}));
    }
    sort(qr.begin(), qr.end());
    int left = 1, right = 0;
    for (Query q : qr) {
        int l = q.l, r = q.r;
        while(right < r) add(++right);
        while(left > l) add(--left);
        while(left < l) Remove(left++);

        while(right > r) Remove(right--);
        ans[q.id] = res;
    }
    for (int i = 0; i < ii; i++) cout << ans[i] << "\n";
}

```

3.7 sparse

```

//__builtin_clz() -> number of leading 0bits from msb
//__builtin_ctz() -> number of leading 0bits from lsb
const int B = 25;
const int N = 1e5 + 10;
vector<int> a(N);
vector<vector<int>> t(B, vector<int>(N, 0));
void build(int n) {
    for (int i = 1; i <= n; ++i) t[0][i] = a[i];
    for (int j = 1; j < B; ++j) for (int i = 1; i + (1 << j) - 1 <= n; ++i) {
        t[j][i] = __gcd(t[j - 1][i], t[j - 1][i + (1 << j) - 1]);
    }
}
int query(int l, int r) {
    int lg = 31 - __builtin_clz(r - l + 1);
    return __gcd(t[lg][l], t[lg][r - (1 << lg) + 1]);
}

```

3.8 trie

```

const int B = 31;
using ll = long long;
struct node {
    int sz;
    node *nxt[2];

    node() {
        nxt[0] = nxt[1] = NULL;
        sz = 0;
    }
} *root;
void del(node* cur) {
    for (int i = 0; i < 2; i++) if (cur -> nxt[i]) del(cur -> nxt[i]);
    delete(cur);
}
void insert(int val) {
    node *cur = root;
    cur -> sz++;

    for (int i = B - 1; i >= 0; --i) {
        int b = val >> i & 1;
        if(cur -> nxt[b] == NULL) cur -> nxt[b] = new node();

        cur = cur -> nxt[b];
        cur -> sz++;
    }
}
void erase(int val) {
    node *cur = root;
    cur -> sz--;

    for (int i = B - 1; i >= 0; --i) {
        int b = val >> i & 1;
        node *tmp = cur;
        cur = cur -> nxt[b];
        cur -> sz--;
        if((cur -> sz) == 0) {
            del(cur);
            tmp -> nxt[b] = NULL; break;
        }
    }
}
int query_max(int val) {
    //return maximum of val ^ x(x is inserted)
}

```

```

int ans = 0;
node *cur = root;

for (int i = B - 1; i >= 0; --i) {
    int b = val >> i & 1;
    if (cur -> nxt[!b] != NULL) ans += (1 << i), cur = cur
        -> nxt[!b];
    else cur = cur -> nxt[b];
}
return ans;
}

int query_min(int val) {
    ///return minimum of val ^ x(x is inserted)
    int ans = 0;
    node *cur = root;

    for (int i = B - 1; i >= 0; --i) {
        int b = val >> i & 1;
        if (cur -> nxt[b] != NULL) cur = cur -> nxt[b];
        else ans += (1 << i), cur = cur -> nxt[!b];
    }
    return ans;
}

int query(ll val, ll k) {
    ///return number of values x, val ^ x < k(x is inserted)
    int ans = 0;
    node *cur = root;
    for (ll i = B - 1; i >= 0; --i) {
        int bit1 = val >> i & 1;
        int bit2 = k >> i & 1;

        if (bit2 == 0) {
            if (cur -> nxt[bit1] != NULL) cur = cur -> nxt[
                bit1];
            else break;
        } else {
            if (cur -> nxt[bit1] != NULL) ans += (cur -> nxt[
                bit1]) -> sz;
            if (cur -> nxt[!bit1] != NULL) cur = cur -> nxt[!
                bit1];
            else break;
        }
    }
    return ans;
}

```

3.9 xorBasis

```

struct XorBasis {

```

```

vector<ll> basis;
ll N = 0, tmp = 0;
void add(ll x) {
    N++;
    tmp ^= x;
    for (ll b : basis) x = min(x, x ^ b);
    if (!x) return;
    for (ll &b : basis) if ((b ^ x) < b) b ^= x;
    basis.push_back(x);
    sort(basis.rbegin(), basis.rend());
}

// number of basis vectors (rank)
ll size() { return (ll) basis.size(); }
// clear everything
void clear() { N = 0; tmp = 0; basis.clear(); }
// returns true if x can be formed
bool possible(ll x) {
    for (ll b : basis) x = min(x, x ^ b);
    return x == 0;
}

// returns maximum xor value
ll maxxor(ll x = 0) {
    for (ll b : basis) x = max(x, x ^ b);
    return x;
}

// returns minimum xor value
ll minxor(ll x = 0) {
    for (ll b : basis) x = min(x, x ^ b);
    return x;
}

// returns count of subsets which xor to x (modded)
ll cntxor(ll x) {
    if (!possible(x)) return 0LL;
    ll freebits = N - size();
    ll res = 1;
    while (freebits--) res = (res * 2) % MOD;
    return res;
}

// returns sum of xor of all subsets
ll sumOfAll() {
    return tmp * (1LL << (N - 1));
}

// returns k-th smallest xor value (1-indexed)
ll kth(ll k) {
    ll sz = size();
    if (k > (1LL << sz)) return -1;
    k--;
    ll ans = 0;
    for (ll i = 0; i < sz; i++)
        if (k >> (sz - 1 - i) & 1) ans ^= basis[i];
}

```

```

        return ans;
    }
} xb;

```

4 Flow

4.1 Dinic

```

struct FlowEdge {
    int v, u;
    int cap, flow = 0;
    FlowEdge(int v, int u, int cap) : v(v), u(u), cap(cap) {}
};

struct Dinic {
    const int flow_inf = 1e8;
    vector<FlowEdge> edges;
    vector<vector<int>> adj;
    int n, m = 0;
    int s, t;
    vector<int> level, ptr;
    queue<int> q;
    Dinic(int n, int s, int t) : n(n), s(s), t(t) {
        adj.resize(n);
        level.resize(n);
        ptr.resize(n);
    }

    void add_edge(int v, int u, int cap) {
        edges.emplace_back(v, u, cap);
        edges.emplace_back(u, v, cap); // cap if undirected,
            otherwise 0;
        adj[v].push_back(m);
        adj[u].push_back(m + 1);
        m += 2;
    }

    bool bfs() {
        while (!q.empty()) {
            int v = q.front();
            q.pop();
            for (int id : adj[v]) {
                if (edges[id].cap == edges[id].flow)
                    continue;
                if (level[edges[id].u] != -1)
                    continue;
                level[edges[id].u] = level[v] + 1;
                q.push(edges[id].u);
            }
        }
        return level[t] != -1;
    }
}

```

```

}
int dfs(int v, int pushed) {
    if (pushed == 0) return 0;
    if (v == t) return pushed;
    for (int& cid = ptr[v]; cid < (int)adj[v].size(); cid++) {
        int id = adj[v][cid];
        int u = edges[id].u;
        if (level[v] + 1 != level[u]) continue;
        int tr = dfs(u, min(pushed, edges[id].cap - edges[id].flow));
        if (tr == 0) continue;
        edges[id].flow += tr; edges[id ^ 1].flow -= tr;
        return tr;
    }
    return 0;
}
int flow() {
    int f = 0;
    while (true) {
        fill(level.begin(), level.end(), -1);
        level[s] = 0;
        q.push(s);
        if (!bfs()) break;
        fill(ptr.begin(), ptr.end(), 0);
        while (int pushed = dfs(s, flow_inf)) {
            f += pushed;
        }
    }
    return f;
}
vector<int> min_cut() {
    vector<int> vis(n, 0);
    queue<int> q;
    q.push(s);
    vis[s] = 1;
    while (!q.empty()) {
        int v = q.front();
        q.pop();
        for (int id : adj[v]) {
            int u = edges[id].u;
            if (!vis[u] && edges[id].cap > edges[id].flow) {
                vis[u] = 1;
                q.push(u);
            }
        }
    }
    return vis; // vis[v] = 1 S-side of cut
}

```

```

}
};

```

4.2 edmondkarp

```

// edmonds karp ve^2 call while(bfs(n) != 0)
typedef struct node {
    int vis; int p; vector<int> adj; vector<ll> cap;
} node;
node graph[519];
ll bfs(int n) {
    queue<int> qu;
    vector<bool> vis(510, 0);
    vis[1] = 1;
    qu.push(1);
    while (!qu.empty()) {
        int cur = qu.front();
        // printf("cur-- %d\n", cur);
        qu.pop();
        f(i, graph[cur].adj.size()) {
            int nex = graph[cur].adj[i];
            if (vis[nex] == 0 && graph[cur].cap[nex] > 0) {
                graph[nex].p = cur; vis[nex] = 1;
                if (nex == n) break;
                qu.push(nex);
            }
        }
    }
    if (vis[n] == 0) return 0;
    ll maxflow = LLONG_MAX;
    int cur = n;
    while (cur != 1) {
        maxflow = min(maxflow, graph[graph[cur].p].cap[cur]);
        cur = graph[cur].p;
        // printf("%d ", cur);
    }
    // printf("max %d\n", maxflow);
    cur = n;
    while (cur != 1) {
        graph[graph[cur].p].cap[cur] -= maxflow;
        graph[cur].cap[graph[cur].p] += maxflow;
        cur = graph[cur].p;
    }
    return maxflow;
}
// call after getting max flow
vector<int> getCut(int n) {

```

```

vector<int> vis(510, 0);
queue<int> q;
q.push(1);
vis[1] = 1;
while (!q.empty()) {
    int u = q.front(); q.pop();
    for (int v : graph[u].adj) {
        if (!vis[v] && graph[u].cap[v] > 0) { // residual capacity
            vis[v] = 1;
            q.push(v);
        }
    }
}
return vis; // vis[v] = 1 -> reachable -> S-side
}

```

5 formulas

```

/* ----- POLYGON -----
Area of polygon (shoelace):
Area = 0.5 * |sum(x[i]*y[i+1] - x[i+1]*y[i])|
Centroid of polygon:
Cx = sum((xi + xi+1) * cross(i, i+1)) / (6*Area)
Cy = sum((yi + yi+1) * cross(i, i+1)) / (6*Area)
Point inside polygon (ray casting):
Count intersections of ray to +x
Odd = inside, Even = outside
Convex polygon point inside:
For all edges (Pi, Pi+1):
cross(Pi+1-Pi, X-Pi) >= 0 (or <=0 consistently)
*/
/* ----- REGULAR POLYGON -----
n = number of sides, s = side length
PI = acos(-1.0)

```

```

Angles (in radians):
Sum of interior = (n - 2) * PI
Single interior = (n - 2) * PI / n
Central / Exterior = 2 * PI / n

Apothem (a) [Inradius - center to midpoint of side]:
a = s / (2 * tan(PI / n))

```

```

Circumradius (R) [Center to vertex]:
R = s / (2 * sin(PI / n))

```

```

Area formulas:

```

```

1. From side: Area = (n * s * s) / (4 * tan(PI / n))
2. From R: Area = 0.5 * n * R * R * sin(2 * PI / n)
3. From Apo: Area = n * s * a * 0.5
*/
/* ----- TRIANGLE -----
Area using cross:
Area = |cross(B-A, C-A)| / 2
Circumcenter:
Intersection of perpendicular bisectors
(Omitted formula compute via lines)
Incenter:
I = (a*A + b*B + c*C) / (a+b+c)
where:
Centroid:
G = (A+B+C)/3
Orthocenter:
H = A + B + C - 2*O (O = circumcenter)
Radius:
Inradius r = Area / s
Circumradius R = (a*b*c) / (4*Area)
Angle between vectors:
cos(theta) = dot(u,v)/(|u||v|)
*/

```

6 Geometry

6.1 Convex hull

```

vector<PT> convex_hull(vector<PT> &p) {
    if (p.size() <= 1) return p;
    vector<PT> v = p;
    sort(v.begin(), v.end());
    vector<PT> up, dn;
    for (auto& p : v) {
        while (up.size() > 1 && orientation(up[up.size() - 2], up.back(), p) >= 0) {
            up.pop_back();
        }
        while (dn.size() > 1 && orientation(dn[dn.size() - 2], dn.back(), p) <= 0) {
            dn.pop_back();
        }
        up.push_back(p);
        dn.push_back(p);
    }
    v = dn;
    if (v.size() > 1) v.pop_back();
    reverse(up.begin(), up.end());
}

```

```

up.pop_back();
for (auto& p : up) {
    v.push_back(p);
}
if (v.size() == 2 && v[0] == v[1]) v.pop_back();
return v;
}
//checks if convex or not
bool is_convex(vector<PT> &p) {
    bool s[3]; s[0] = s[1] = s[2] = 0;
    int n = p.size();
    for (int i = 0; i < n; i++) {
        int j = (i + 1) % n;
        int k = (j + 1) % n;
        s[sign(cross(p[j] - p[i], p[k] - p[i])) + 1] = 1;
        if (s[0] && s[2]) return 0;
    }
    return 1;
}

```

6.2 Geo 2

```

struct point { int x, y, idx; }; //Finds squared euclidean
distance between two points int dist(point &a, point &b) {
    return (a.x-b.x)*(a.x-b.x) + (a.y-b.y)*(a.y-b.y); }
//Checks if angle ABC is a right angle int
isOrthogonal(point &a, point &b, point &c) { return (b.x-a.x) * (b.x-c.x) + (b.y-a.y) * (b.y-c.y) == 0; } //
Checks if ABCD form a rectangle (in that order) int
isRectangle(point &a, point &b, point &c, point &d) {
    return isOrthogonal(a, b, c) && isOrthogonal(b, c, d)
    && isOrthogonal(c, d, a); } //Checks if ABCD form a
rectangle, in any orientation int isRectangleAnyOrder(
point &a, point &b, point &c, point &d) { return
isRectangle(a, b, c, d) || isRectangle(b, c, a, d) |
isRectangle(c, a, b, d); } //Checks if ABCD form a
square (in that order) int isSquare(point &a, point &b,
point &c, point &d) { return isRectangle(a, b, c, d)
&& dist(a, b) == dist(b, c); } //Checks if ABCD form a
square, in any orientation int isSquareAnyOrder(point &
a, point &b, point &c, point &d) { return isSquare(a, b,
c, d) || isSquare(b, c, a, d) | isSquare(c, a, b, d); }
}

```

6.3 geo_{2d}

```

const double PI = acos((double)-1.0); int sign(double x) {
    return (x > eps) - (x < -eps); } struct PT { double x,
y; PT() { x = 0, y = 0; } PT(double x, double y) : x(x),
y(y) {} PT(const PT &p) : x(p.x), y(p.y) {} PT
operator + (const PT &a) const { return PT(x + a.x, y +
a.y); } PT operator - (const PT &a) const { return PT(
x - a.x, y - a.y); } PT operator * (const double a)
const { return PT(x * a, y * a); } friend PT operator *
(const double &a, const PT &b) { return PT(a * b.x, a
* b.y); } PT operator / (const double a) const { return
PT(x / a, y / a); } bool operator == (PT a) const {
    return sign(a.x - x) == 0 && sign(a.y - y) == 0; } bool
operator != (PT a) const { return !(this == a); }
bool operator < (PT a) const { return sign(a.x - x) ==
0 ? y < a.y : x < a.x; } bool operator > (PT a) const {
    return sign(a.x - x) == 0 ? y > a.y : x > a.x; }
double norm() { return sqrt(x * x + y * y); } double
norm2() { return x * x + y * y; } PT perp() { return PT
(-y, x); } double arg() { return atan2(y, x); } PT
truncate(double r) { // returns a vector with norm r
and having same direction double k = norm(); if (!sign(
k)) return *this; r /= k; return PT(x * r, y * r); } };
istream &operator >> (istream &in, PT &p) { return in >> p.x
>> p.y; }
ostream &operator << (ostream &out, PT &p) { return out << "
(" << p.x << ", " << p.y << ")"; }
inline double dot(PT a, PT b) { return a.x * b.x + a.y * b.y
; }
inline double dist2(PT a, PT b) { return dot(a - b, a - b);
}
inline double dist(PT a, PT b) { return sqrt(dot(a - b, a -
b)); }
inline double cross(PT a, PT b) { return a.x * b.y - a.y * b
.x; }
inline double cross2(PT a, PT b, PT c) { return cross(b - a,
c - a); }
inline int orientation(PT a, PT b, PT c) { return sign(cross
(b - a, c - a)); }
PT perp(PT a) { return PT(-a.y, a.x); }
PT rotateccw90(PT a) { return PT(-a.y, a.x); }
PT rotatecw90(PT a) { return PT(a.y, -a.x); }
PT rotateccw(PT a, double t) { return PT(a.x * cos(t) - a.y
* sin(t), a.x * sin(t) + a.y * cos(t)); }
PT rotatecw(PT a, double t) { return PT(a.x * cos(t) + a.y *
sin(t), -a.x * sin(t) + a.y * cos(t)); }
double SQ(double x) { return x * x; }
double rad_to_deg(double r) { return (r * 180.0 / PI); }
double deg_to_rad(double d) { return (d * PI / 180.0); }
double get_angle(PT a, PT b) {
    double costheta = dot(a, b) / a.norm() / b.norm();
}

```

```

    return acos(max((double)-1.0, min((double)1.0, costheta))
    );
}
bool is_point_in_angle(PT b, PT a, PT c, PT p) { // does
    point p lie in angle <bac
    assert(orientation(a, b, c) != 0);
    if (orientation(a, c, b) < 0) swap(b, c);
    return orientation(a, c, p) >= 0 && orientation(a, b, p)
    <= 0;
}
bool half(PT p) {
    return p.y > 0.0 || (p.y == 0.0 && p.x < 0.0);
}
void polar_sort(vector<PT> &v) { // sort points in
    counterclockwise
    sort(v.begin(), v.end(), [](PT a, PT b) {
        return make_tuple(half(a), 0.0, a.norm2()) <
        make_tuple(half(b), cross(a, b), b.norm2());
    });
}
void polar_sort(vector<PT> &v, PT o) { // sort points in
    counterclockwise with respect to point o
    sort(v.begin(), v.end(), [&](PT a, PT b) {
        return make_tuple(half(a - o), 0.0, (a - o).norm2())
        < make_tuple(half(b - o), cross(a - o, b - o), (
        b - o).norm2());
    });
}
struct line {
    PT a, b; // goes through points a and b
    PT v; double c; //line form: direction vec [cross] (x, y)
    = c
    line() {}
    //direction vector v and offset c
    line(PT v, double c) : v(v), c(c) {
        auto p = get_points();
        a = p.first; b = p.second;
    }
    // equation ax + by + c = 0
    line(double _a, double _b, double _c) : v({_b, -_a}), c(-_c
    ) {
        auto p = get_points();
        a = p.first; b = p.second;
    }
    // goes through points p and q
    line(PT p, PT q) : v(q - p), c(cross(v, p)), a(p), b(q) {}
    pair<PT, PT> get_points() { //extract any two points
        from this line
    }
    PT p, q; double a = -v.y, b = v.x; // ax + by = c
    if (sign(a) == 0) {

```

```

        p = PT(0, c / b); q = PT(1, c / b);
    }
    else if (sign(b) == 0) {
        p = PT(c / a, 0); q = PT(c / a, 1);
    }
    else {
        p = PT(0, c / b); q = PT(1, (c - a) / b);
    }
    return {p, q};
    }
    // ax + by + c = 0
    array<double, 3> get_abc() {
        double a = -v.y, b = v.x; return {a, b, -c};
    }
    // 1 if on the left, -1 if on the right, 0 if on the line
    int side(PT p) { return sign(cross(v, p) - c); }
    // line that is perpendicular to this and goes through
    point p
    line perpendicular_through(PT p) { return {p, p + perp(v)
    }; }
    // translate the line by vector t i.e. shifting it by
    vector t
    line translate(PT t) { return {v, c + cross(v, t)}; }
    // compare two points by their orthogonal projection on
    this line
    // a projection point comes before another if it comes
    first according to vector v
    bool cmp_by_projection(PT p, PT q) { return dot(v, p) <
    dot(v, q); }
    line shift_left(double d) {
        PT z = v.perp().truncate(d);
        return line(a + z, b + z);
    }
};
// find a point from a through b with distance d
PT point_along_line(PT a, PT b, double d) {
    assert(a != b);
    return a + (((b - a) / (b - a).norm()) * d);
}
// projection point c onto line through a and b assuming a
!= b
PT project_from_point_to_line(PT a, PT b, PT c) {
    return a + (b - a) * dot(c - a, b - a) / (b - a).norm2();
}
// reflection point c onto line through a and b assuming a
!= b
PT reflection_from_point_to_line(PT a, PT b, PT c) {
    PT p = project_from_point_to_line(a, b, c);
    return p + p - c;
}

```

```

// minimum distance from point c to line through a and b
double dist_from_point_to_line(PT a, PT b, PT c) {
    return fabs(cross(b - a, c - a) / (b - a).norm());
}
// returns true if point p is on line segment ab
bool is_point_on_seg(PT a, PT b, PT p) {
    if (fabs(cross(p - b, a - b)) < eps) {
        if (p.x < min(a.x, b.x) - eps || p.x > max(a.x, b.x)
        + eps) return false;
        if (p.y < min(a.y, b.y) - eps || p.y > max(a.y, b.y)
        + eps) return false;
        return true;
    }
    return false;
}
// minimum distance point from point c to segment ab that
lies on segment ab
PT project_from_point_to_seg(PT a, PT b, PT c) {
    double r = dist2(a, b);
    if (sign(r) == 0) return a;
    r = dot(c - a, b - a) / r;
    if (r < 0) return a;
    if (r > 1) return b;
    return a + (b - a) * r;
}
// minimum distance from point c to segment ab
double dist_from_point_to_seg(PT a, PT b, PT c) {
    return dist(c, project_from_point_to_seg(a, b, c));
}
// 0 if not parallel, 1 if parallel, 2 if collinear
int is_parallel(PT a, PT b, PT c, PT d) {
    double k = fabs(cross(b - a, d - c));
    if (k < eps) {
        if (fabs(cross(a - b, a - c)) < eps && fabs(cross(c -
        d, c - a)) < eps) return 2;
        else return 1;
    }
    else return 0;
}
// check if two lines are same
bool are_lines_same(PT a, PT b, PT c, PT d) {
    if (fabs(cross(a - c, c - d)) < eps && fabs(cross(b - c,
    c - d)) < eps) return true;
    return false;
}
// bisector vector of <abc
PT angle_bisector(PT &a, PT &b, PT &c) {
    PT p = a - b, q = c - b;
    return p + q * sqrt(dot(p, p) / dot(q, q));
}

```



```

// 1 if point is ccw to the line, 2 if point is cw to the
// line, 3 if point is on the line
int point_line_relation(PT a, PT b, PT p) {
    int c = sign(cross(p - a, b - a));
    if (c < 0) return 1;
    if (c > 0) return 2;
    return 3;
}
// intersection point between ab and cd assuming unique
// intersection exists
bool line_line_intersection(PT a, PT b, PT c, PT d, PT &ans)
{
    double a1 = a.y - b.y, b1 = b.x - a.x, c1 = cross(a, b);
    double a2 = c.y - d.y, b2 = d.x - c.x, c2 = cross(c, d);
    double det = a1 * b2 - a2 * b1;
    if (det == 0) return 0;
    ans = PT((b1 * c2 - b2 * c1) / det, (c1 * a2 - a1 * c2) /
    det);
    return 1;
}
// intersection point between segment ab and segment cd
// assuming unique intersection exists
bool seg_seg_intersection(PT a, PT b, PT c, PT d, PT &ans) {
    double oa = cross2(c, d, a), ob = cross2(c, d, b);
    double oc = cross2(a, b, c), od = cross2(a, b, d);
    if (oa * ob < 0 && oc * od < 0) {
        ans = (a * ob - b * oa) / (ob - oa);
        return 1;
    }
    else return 0;
}
// intersection point between segment ab and segment cd
// assuming unique intersection may not exists
// se.size()==0 means no intersection
// se.size()==1 means one intersection
// se.size()==2 means range intersection
set<PT> seg_seg_intersection_inside(PT a, PT b, PT c, PT d)
{
    PT ans;
    if (seg_seg_intersection(a, b, c, d, ans)) return {ans};
    set<PT> se;
    if (is_point_on_seg(c, d, a)) se.insert(a);
    if (is_point_on_seg(c, d, b)) se.insert(b);
    if (is_point_on_seg(a, b, c)) se.insert(c);
    if (is_point_on_seg(a, b, d)) se.insert(d);
    return se;
}
// intersection between segment ab and line cd
// 0 if do not intersect, 1 if proper intersect, 2 if
// segment intersect

```

```

int seg_line_relation(PT a, PT b, PT c, PT d) {
    double p = cross2(c, d, a);
    double q = cross2(c, d, b);
    if (sign(p) == 0 && sign(q) == 0) return 2;
    else if (p * q < 0) return 1;
    else return 0;
}
// intersection between segment ab and line cd assuming
// unique intersection exists
bool seg_line_intersection(PT a, PT b, PT c, PT d, PT &ans)
{
    bool k = seg_line_relation(a, b, c, d);
    assert(k != 2);
    if (k) line_line_intersection(a, b, c, d, ans);
    return k;
}
// minimum distance from segment ab to segment cd
double dist_from_seg_to_seg(PT a, PT b, PT c, PT d) {
    PT dummy;
    if (seg_seg_intersection(a, b, c, d, dummy)) return 0.0;
    else return min({dist_from_point_to_seg(a, b, c),
    dist_from_point_to_seg(a, b, d),
    dist_from_point_to_seg(c, d, a),
    dist_from_point_to_seg(c, d, b)});
}
// minimum distance from point c to ray (starting point a
// and direction vector b)
double dist_from_point_to_ray(PT a, PT b, PT c) {
    b = a + b;
    double r = dot(c - a, b - a);
    if (r < 0.0) return dist(c, a);
    return dist_from_point_to_line(a, b, c);
}
// starting point as and direction vector ad
bool ray_ray_intersection(PT as, PT ad, PT bs, PT bd) {
    double dx = bs.x - as.x, dy = bs.y - as.y;
    double det = bd.x * ad.y - bd.y * ad.x;
    if (fabs(det) < eps) return 0;
    double u = (dy * bd.x - dx * bd.y) / det;
    double v = (dy * ad.x - dx * ad.y) / det;
    if (sign(u) >= 0 && sign(v) >= 0) return 1;
    else return 0;
}
double ray_ray_distance(PT as, PT ad, PT bs, PT bd) {
    if (ray_ray_intersection(as, ad, bs, bd)) return 0.0;
    double ans = dist_from_point_to_ray(as, ad, bs);
    ans = min(ans, dist_from_point_to_ray(bs, bd, as));
    return ans;
}
struct circle {

```

```

PT p; double r;
circle() {}
circle(PT _p, double _r): p(_p), r(_r) {};
// center (x, y) and radius r
circle(double x, double y, double _r): p(PT(x, y)), r(_r)
{};
// circumcircle of a triangle
// the three points must be unique
circle(PT a, PT b, PT c) {
    b = (a + b) * 0.5;
    c = (a + c) * 0.5;
    line_line_intersection(b, b + rotatecw90(a - b), c, c
    + rotatecw90(a - c), p);
    r = dist(a, p);
}
// inscribed circle of a triangle
// pass a bool just to differentiate from circumcircle
circle(PT a, PT b, PT c, bool t) {
    line u, v;
    double m = atan2(b.y - a.y, b.x - a.x), n = atan2(c.y
    - a.y, c.x - a.x);
    u.a = a;
    u.b = u.a + (PT(cos((n + m)/2.0), sin((n + m)/2.0)));
    v.a = b;
    m = atan2(a.y - b.y, a.x - b.x), n = atan2(c.y - b.y,
    c.x - b.x);
    v.b = v.a + (PT(cos((n + m)/2.0), sin((n + m)/2.0)));
    line_line_intersection(u.a, u.b, v.a, v.b, p);
    r = dist_from_point_to_seg(a, b, p);
}
bool operator == (circle v) { return p == v.p && sign(r -
    v.r) == 0; }
double area() { return PI * r * r; }
double circumference() { return 2.0 * PI * r; }
};
//0 if outside, 1 if on circumference, 2 if inside circle
int circle_point_relation(PT p, double r, PT b) {
    double d = dist(p, b);
    if (sign(d - r) < 0) return 2;
    if (sign(d - r) == 0) return 1;
    return 0;
}
// 0 if outside, 1 if on circumference, 2 if inside circle
int circle_line_relation(PT p, double r, PT a, PT b) {
    double d = dist_from_point_to_line(a, b, p);
    if (sign(d - r) < 0) return 2;
    if (sign(d - r) == 0) return 1;
    return 0;
}
//compute intersection of line through points a and b with

```

```

//circle centered at c with radius r > 0
vector<PT> circle_line_intersection(PT c, double r, PT a, PT
    b) {
    vector<PT> ret;
    b = b - a; a = a - c;
    double A = dot(b, b), B = dot(a, b);
    double C = dot(a, a) - r * r, D = B * B - A * C;
    if (D < -eps) return ret;
    ret.push_back(c + a + b * (-B + sqrt(D + eps)) / A);
    if (D > eps) ret.push_back(c + a + b * (-B - sqrt(D)) / A
    );
    return ret;
}
//5 - outside and do not intersect
//4 - intersect outside in one point
//3 - intersect in 2 points
//2 - intersect inside in one point
//1 - inside and do not intersect
int circle_circle_relation(PT a, double r, PT b, double R) {
    double d = dist(a, b);
    if (sign(d - r - R) > 0) return 5;
    if (sign(d - r - R) == 0) return 4;
    double l = fabs(r - R);
    if (sign(d - r - R) < 0 && sign(d - l) > 0) return 3;
    if (sign(d - l) == 0) return 2;
    if (sign(d - l) < 0) return 1;
    assert(0); return -1;
}
vector<PT> circle_circle_intersection(PT a, double r, PT b,
    double R) {
    if (a == b && sign(r - R) == 0) return {PT(1e18, 1e18)};
    vector<PT> ret;
    double d = sqrt(dist2(a, b));
    if (d > r + R || d + min(r, R) < max(r, R)) return ret;
    double x = (d * d - R * R + r * r) / (2 * d);
    double y = sqrt(r * r - x * x);
    PT v = (b - a) / d;
    ret.push_back(a + v * x + rotateccw90(v) * y);
    if (y > 0) ret.push_back(a + v * x - rotateccw90(v) * y);
    return ret;
}
// returns two circle c1, c2 through points a, b and of
// radius r
// 0 if there is no such circle, 1 if one circle, 2 if two
// circle
int get_circle(PT a, PT b, double r, circle &c1, circle &c2)
{
    vector<PT> v = circle_circle_intersection(a, r, b, r);
    int t = v.size();
    if (!t) return 0;

```

```

    c1.p = v[0], c1.r = r;
    if (t == 2) c2.p = v[1], c2.r = r;
    return t;
}
// returns two circle c1, c2 which is tangent to line u,
// goes through
// point q and has radius r1; 0 for no circle, 1 if c1 = c2
// , 2 if c1 != c2
int get_circle(line u, PT q, double r1, circle &c1, circle &
    c2) {
    double d = dist_from_point_to_line(u.a, u.b, q);
    if (sign(d - r1 * 2.0) > 0) return 0;
    if (sign(d) == 0) {
        cout << u.v.x << ' ' << u.v.y << '\n';
        c1.p = q + rotateccw90(u.v).truncate(r1);
        c2.p = q + rotatecw90(u.v).truncate(r1);
        c1.r = c2.r = r1;
        return 2;
    }
    line u1 = line(u.a + rotateccw90(u.v).truncate(r1), u.b +
        rotateccw90(u.v).truncate(r1));
    line u2 = line(u.a + rotatecw90(u.v).truncate(r1), u.b +
        rotatecw90(u.v).truncate(r1));
    circle cc = circle(q, r1);
    PT p1, p2; vector<PT> v;
    v = circle_line_intersection(q, r1, u1.a, u1.b);
    if (!v.size()) v = circle_line_intersection(q, r1, u2.a,
        u2.b);
    v.push_back(v[0]);
    p1 = v[0], p2 = v[1];
    c1 = circle(p1, r1);
    if (p1 == p2) {
        c2 = c1;
        return 1;
    }
    c2 = circle(p2, r1);
    return 2;
}
// returns area of intersection between two circles
double circle_circle_area(PT a, double r1, PT b, double r2)
{
    double d = (a - b).norm();
    if (r1 + r2 < d + eps) return 0;
    if (r1 + d < r2 + eps) return PI * r1 * r1;
    if (r2 + d < r1 + eps) return PI * r2 * r2;
    double theta_1 = acos((r1 * r1 + d * d - r2 * r2) / (2 *
        r1 * d));
    theta_2 = acos((r2 * r2 + d * d - r1 * r1) / (2 * r2 * d));
    ;

```

```

    return r1 * r1 * (theta_1 - sin(2 * theta_1)/2.) + r2 *
        r2 * (theta_2 - sin(2 * theta_2)/2.);
}
// tangent lines from point q to the circle
int tangent_lines_from_point(PT p, double r, PT q, line &u,
    line &v) {
    int x = sign(dist2(p, q) - r * r);
    if (x < 0) return 0; // point in circle
    if (x == 0) { // point on circle
        u = line(q, q + rotateccw90(q - p));
        v = u;
        return 1;
    }
    double d = dist(p, q);
    double l = r * r / d;
    double h = sqrt(r * r - l * l);
    u = line(q, p + ((q - p).truncate(l) + (rotateccw90(q - p
        ).truncate(h))));
    v = line(q, p + ((q - p).truncate(l) + (rotatecw90(q - p)
        .truncate(h))));
    return 2;
}
// returns outer tangents line of two circles
// if inner == 1 it returns inner tangent lines
int tangents_lines_from_circle(PT c1, double r1, PT c2,
    double r2, bool inner, line &u, line &v) {
    if (inner) r2 = -r2;
    PT d = c2 - c1;
    double dr = r1 - r2, d2 = d.norm2(), h2 = d2 - dr * dr;
    if (d2 == 0 || h2 < 0) {
        assert(h2 != 0);
        return 0;
    }
    vector<pair<PT, PT>>out;
    for (int tmp: {-1, 1}) {
        PT v = (d * dr + rotateccw90(d) * sqrt(h2) * tmp) /
            d2;
        out.push_back({c1 + v * r1, c2 + v * r2});
    }
    u = line(out[0].first, out[0].second);
    if (out.size() == 2) v = line(out[1].first, out[1].second
        );
    return 1 + (h2 > 0);
}
double area_of_triangle(PT a, PT b, PT c) {
    return fabs(cross(b - a, c - a) * 0.5);
}

```

```

// -1 if strictly inside, 0 if on the polygon, 1 if strictly
// outside
int is_point_in_triangle(PT a, PT b, PT c, PT p) {
    if (sign(cross(b - a, c - a)) < 0) swap(b, c);
    int c1 = sign(cross(b - a, p - a));
    int c2 = sign(cross(c - b, p - b));
    int c3 = sign(cross(a - c, p - c));
    if (c1 < 0 || c2 < 0 || c3 < 0) return 1;
    if (c1 + c2 + c3 != 3) return 0;
    return -1;
}

double perimeter(vector<PT> &p) {
    double ans = 0; int n = p.size();
    for (int i = 0; i < n; i++) ans += dist(p[i], p[(i + 1) % n]);
    return ans;
}

double area(vector<PT> &p) {
    double ans = 0; int n = p.size();
    for (int i = 0; i < n; i++) ans += cross(p[i], p[(i + 1) % n]);
    return fabs(ans) * 0.5;
}

// centroid of a (possibly non-convex) polygon,
// assuming that the coordinates are listed in a clockwise
// or
// counterclockwise fashion. Note that the centroid is often
// known as
// the "center of gravity" or "center of mass".
PT centroid(vector<PT> &p) {
    int n = p.size(); PT c(0, 0);
    double sum = 0;
    for (int i = 0; i < n; i++) sum += cross(p[i], p[(i + 1) % n]);
    double scale = 3.0 * sum;
    for (int i = 0; i < n; i++) {
        int j = (i + 1) % n;
        c = c + (p[i] + p[j]) * cross(p[i], p[j]);
    }
    return c / scale;
}

// 0 if cw, 1 if ccw
bool get_direction(vector<PT> &p) {
    double ans = 0; int n = p.size();
    for (int i = 0; i < n; i++) ans += cross(p[i], p[(i + 1) % n]);
    if (sign(ans) > 0) return 1;
    return 0;
}

// it returns a point such that the sum of distances

```

```

// from that point to all points in p is minimum
// 0(n log^2 MX)
PT geometric_median(vector<PT> p) {
    auto tot_dist = [&](PT z) {
        double res = 0;
        for (int i = 0; i < p.size(); i++) res += dist(p[i], z);
        return res;
    };
    auto findY = [&](double x) {
        double y1 = -1e5, yr = 1e5;
        for (int i = 0; i < 60; i++) {
            double ym1 = y1 + (yr - y1) / 3;
            double ym2 = yr - (yr - y1) / 3;
            double d1 = tot_dist(PT(x, ym1));
            double d2 = tot_dist(PT(x, ym2));
            if (d1 < d2) yr = ym2;
            else y1 = ym1;
        }
        return pair<double, double> (y1, tot_dist(PT(x, y1)));
    };
    double x1 = -1e5, xr = 1e5;
    for (int i = 0; i < 60; i++) {
        double xm1 = x1 + (xr - x1) / 3;
        double xm2 = xr - (xr - x1) / 3;
        double y1, d1, y2, d2;
        auto z = findY(xm1); y1 = z.first; d1 = z.second;
        z = findY(xm2); y2 = z.first; d2 = z.second;
        if (d1 < d2) xr = xm2;
        else x1 = xm1;
    }
    return {x1, findY(x1).first};
}

// -1 if strictly inside, 0 if on the polygon, 1 if strictly
// outside
// it must be strictly convex, otherwise make it strictly
// convex first
int is_point_in_convex(vector<PT> &p, const PT& x) { // 0(
    log n)
    int n = p.size(); assert(n >= 3);
    int a = orientation(p[0], p[1], x), b = orientation(p[0],
        p[n - 1], x);
    if (a < 0 || b > 0) return 1;
    int l = 1, r = n - 1;
    while (l + 1 < r) {
        int mid = l + r >> 1;
        if (orientation(p[0], p[mid], x) >= 0) l = mid;
        else r = mid;
    }
    int k = orientation(p[1], p[r], x);
    if (k <= 0) return -k;
}

```

```

    if (l == 1 && a == 0) return 0;
    if (r == n - 1 && b == 0) return 0;
    return -1;
}

bool is_point_on_polygon(vector<PT> &p, const PT& z) {
    int n = p.size();
    for (int i = 0; i < n; i++) {
        if (is_point_on_seg(p[i], p[(i + 1) % n], z)) return 1;
    }
    return 0;
}

// returns 1e9 if the point is on the polygon
int winding_number(vector<PT> &p, const PT& z) { // 0(n)
    if (is_point_on_polygon(p, z)) return 1e9;
    int n = p.size(), ans = 0;
    for (int i = 0; i < n; ++i) {
        int j = (i + 1) % n;
        bool below = p[i].y < z.y;
        if (below != (p[j].y < z.y)) {
            auto orient = orientation(z, p[j], p[i]);
            if (orient == 0) return 0;
            if (below == (orient > 0)) ans += below ? 1 : -1;
        }
    }
    return ans;
}

// -1 if strictly inside, 0 if on the polygon, 1 if strictly
// outside
int is_point_in_polygon(vector<PT> &p, const PT& z) { // 0(n)
    int k = winding_number(p, z);
    return k == 1e9 ? 0 : k == 0 ? 1 : -1;
}

double diameter(vector<PT> &p) {
    int n = (int)p.size();
    if (n == 1) return 0;
    if (n == 2) return dist(p[0], p[1]);
    double ans = 0;
    int i = 0, j = 1;
    while (i < n) {
        while (cross(p[(i + 1) % n] - p[i], p[(j + 1) % n] -
            p[j]) >= 0) {
            ans = max(ans, dist2(p[i], p[j]));
            j = (j + 1) % n;
        }
        ans = max(ans, dist2(p[i], p[j]));
        i++;
    }
    return sqrt(ans);
}

```

```

}
// minimum distance between two parallel lines (non
// necessarily axis parallel)
// such that the polygon can be put between the lines
double width(vector<PT> &p) {
    int n = (int)p.size();
    if (n <= 2) return 0;
    double ans = inf;
    int i = 0, j = 1;
    while (i < n) {
        while (cross(p[(i + 1) % n] - p[i], p[(j + 1) % n] -
            p[j]) >= 0) j = (j + 1) % n;
        ans = min(ans, dist_from_point_to_line(p[i], p[(i +
            1) % n], p[j]));
        i++;
    }
    return ans;
}
// minimum perimeter
double minimum_enclosing_rectangle(vector<PT> &p) {
    int n = p.size();
    if (n <= 2) return perimeter(p);
    int mndot = 0; double tmp = dot(p[1] - p[0], p[0]);
    for (int i = 1; i < n; i++) {
        if (dot(p[1] - p[0], p[i]) <= tmp) {
            tmp = dot(p[1] - p[0], p[i]);
            mndot = i;
        }
    }
    double ans = inf;
    int i = 0, j = 1, mxdot = 1;
    while (i < n) {
        PT cur = p[(i + 1) % n] - p[i];
        while (cross(cur, p[(j + 1) % n] - p[j]) >= 0) j = (j
            + 1) % n;
        while (dot(p[(mxdot + 1) % n], cur) >= dot(p[mxdot],
            cur)) mxdot = (mxdot + 1) % n;
        while (dot(p[(mndot + 1) % n], cur) <= dot(p[mndot],
            cur)) mndot = (mndot + 1) % n;
        ans = min(ans, 2.0 * ((dot(p[mxdot], cur) / cur.norm
            ()) - dot(p[mndot], cur) / cur.norm()) +
            dist_from_point_to_line(p[i], p[(i + 1) % n], p[
                j])));
        i++;
    }
    return ans;
}
// given n points, find the minimum enclosing circle of the
// points
// call convex_hull() before this for faster solution

```

```

// expected O(n)
circle minimum_enclosing_circle(vector<PT> &p) {
    random_shuffle(p.begin(), p.end());
    int n = p.size();
    circle c(p[0], 0);
    for (int i = 1; i < n; i++) {
        if (sign(dist(c.p, p[i]) - c.r) > 0) {
            c = circle(p[i], 0);
            for (int j = 0; j < i; j++) {
                if (sign(dist(c.p, p[j]) - c.r) > 0) {
                    c = circle((p[i] + p[j]) / 2, dist(p[i], p
                        [j]) / 2);
                    for (int k = 0; k < j; k++) {
                        if (sign(dist(c.p, p[k]) - c.r) > 0) {
                            c = circle(p[i], p[j], p[k]);
                        }
                    }
                }
            }
        }
    }
    return c;
}
pair<PT, int> point_poly_tangent(vector<PT> &p, PT Q, int
    dir, int l, int r) {
    while (r - l > 1) {
        int mid = (l + r) >> 1;
        bool pvs = orientation(Q, p[mid], p[mid - 1]) != -dir
            ;
        bool nxt = orientation(Q, p[mid], p[mid + 1]) != -dir
            ;
        if (pvs && nxt) return {p[mid], mid};
        if (!(pvs || nxt)) {
            auto p1 = point_poly_tangent(p, Q, dir, mid + 1,
                r);
            auto p2 = point_poly_tangent(p, Q, dir, l, mid -
                1);
            return orientation(Q, p1.first, p2.first) == dir
                ? p1 : p2;
        }
        if (!pvs) {
            if (orientation(Q, p[mid], p[l]) == dir) r = mid
                - 1;
            else if (orientation(Q, p[l], p[r]) == dir) r =
                mid - 1;
            else l = mid + 1;
        }
        if (!nxt) {

```

```

            if (orientation(Q, p[mid], p[l]) == dir) l = mid
                + 1;
            else if (orientation(Q, p[l], p[r]) == dir) r =
                mid - 1;
            else l = mid + 1;
        }
    }
    pair<PT, int> ret = {p[l], l};
    for (int i = l + 1; i <= r; i++) ret = orientation(Q, ret
        .first, p[i]) != dir ? make_pair(p[i], i) : ret;
    return ret;
}
// (ccw, cw) tangents from a point that is outside this
// convex polygon
// returns indexes of the points
// ccw means the tangent from Q to that point is in the same
// direction as the polygon ccw direction
pair<int, int> tangents_from_point_to_polygon(vector<PT> &p,
    PT Q) {
    int ccw = point_poly_tangent(p, Q, 1, 0, (int)p.size() -
        1).second;
    int cw = point_poly_tangent(p, Q, -1, 0, (int)p.size() -
        1).second;
    return make_pair(ccw, cw);
}
// minimum distance from a point to a convex polygon
// it assumes point lie strictly outside the polygon
double dist_from_point_to_polygon(vector<PT> &p, PT z) {
    double ans = inf;
    int n = p.size();
    if (n <= 3) {
        for (int i = 0; i < n; i++) ans = min(ans,
            dist_from_point_to_seg(p[i], p[(i + 1) % n], z));
        return ans;
    }
    auto [r, l] = tangents_from_point_to_polygon(p, z);
    if (l > r) r += n;
    while (l < r) {
        int mid = (l + r) >> 1;
        double left = dist2(p[mid % n], z), right = dist2(p[
            (mid + 1) % n], z);
        ans = min({ans, left, right});
        if (left < right) r = mid;
        else l = mid + 1;
    }
    ans = sqrt(ans);
    ans = min(ans, dist_from_point_to_seg(p[l % n], p[(l + 1)
        % n], z));
}

```

```

    ans = min(ans, dist_from_point_to_seg(p[l % n], p[(l - 1 + n) % n], z));
    return ans;
}
// minimum distance from convex polygon p to line ab
// returns 0 is it intersects with the polygon
// top - upper right vertex
double dist_from_polygon_to_line(vector<PT> &p, PT a, PT b,
    int top) { //O(log n)
    PT orth = (b - a).perp();
    if (orientation(a, b, p[0]) > 0) orth = (a - b).perp();
    int id = extreme_vertex(p, orth, top);
    if (dot(p[id] - a, orth) > 0) return 0.0; //if orth and a
    // are in the same half of the line, then poly and line
    // intersects
    return dist_from_point_to_line(a, b, p[id]); //does not
    // intersect
}
// minimum distance from a convex polygon to another convex
// polygon
// the polygon doesnot overlap or touch
// tested in https://toph.co/p/the-wall
double dist_from_polygon_to_polygon(vector<PT> &p1, vector<
    PT> &p2) { // 0(n log n)
    double ans = inf;
    for (int i = 0; i < p1.size(); i++) {
        ans = min(ans, dist_from_point_to_polygon(p2, p1[i]));
    }
    for (int i = 0; i < p2.size(); i++) {
        ans = min(ans, dist_from_point_to_polygon(p1, p2[i]));
    }
    return ans;
}
// maximum distance from a convex polygon to another convex
// polygon
double maximum_dist_from_polygon_to_polygon(vector<PT> &u,
    vector<PT> &v){ //O(n)
    int n = (int)u.size(), m = (int)v.size();
    double ans = 0;
    if (n < 3 || m < 3) {
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < m; j++) ans = max(ans, dist2(
                u[i], v[j]));
        }
        return sqrt(ans);
    }
    if (u[0].x > v[0].x) swap(n, m), swap(u, v);
    int i = 0, j = 0, step = n + m + 10;

```

```

    while (j + 1 < m && v[j].x < v[j + 1].x) j++;
    while (step-- > 0) {
        if (cross(u[(i + 1) % n] - u[i], v[(j + 1) % m] - v[j])
            >= 0) j = (j + 1) % m;
        else i = (i + 1) % n;
        ans = max(ans, dist2(u[i], v[j]));
    }
    return sqrt(ans);
}
// rotate the polygon such that the (bottom, left)-most
// point is at the first position
void reorder_polygon(vector<PT> &p) {
    int pos = 0;
    for (int i = 1; i < p.size(); i++) {
        if (p[i].y < p[pos].y || (sign(p[i].y - p[pos].y) == 0 &&
            p[i].x < p[pos].x)) pos = i;
    }
    rotate(p.begin(), p.begin() + pos, p.end());
}
// returns the area of the intersection of the circle with
// center c and radius r
// and the triangle formed by the points c, a, b
double _triangle_circle_intersection(PT c, double r, PT a,
    PT b) {
    double sd1 = dist2(c, a), sd2 = dist2(c, b);
    if (sd1 > sd2) swap(a, b), swap(sd1, sd2);
    double sd = dist2(a, b);
    double d1 = sqrt1(sd1), d2 = sqrt1(sd2), d = sqrt(sd);
    double x = abs(sd2 - sd - sd1) / (2 * d);
    double h = sqrt1(sd1 - x * x);
    if (r >= d2) return h * d / 2;
    double area = 0;
    if (sd + sd1 < sd2) {
        if (r < d1) area = r * r * (acos(h / d2) - acos(h / d1)) / 2;
        else {
            area = r * r * (acos(h / d2) - acos(h / r)) / 2;
            double y = sqrt1(r * r - h * h);
            area += h * (y - x) / 2;
        }
    }
    else {
        if (r < h) area = r * r * (acos(h / d2) + acos(h / d1)) / 2;
        else {
            area += r * r * (acos(h / d2) - acos(h / r)) / 2;
            double y = sqrt1(r * r - h * h);
            area += h * y / 2;
            if (r < d1) {

```

```

                area += r * r * (acos(h / d1) - acos(h / r)) / 2;
                area += h * y / 2;
            }
        }
        else area += h * x / 2;
    }
}
// intersection between a simple polygon and a circle
double polygon_circle_intersection(vector<PT> &v, PT p,
    double r) {
    int n = v.size();
    double ans = 0.00;
    PT org = {0, 0};
    for (int i = 0; i < n; i++) {
        int x = orientation(p, v[i], v[(i + 1) % n]);
        if (x == 0) continue;
        double area = _triangle_circle_intersection(org, r, v
            [i] - p, v[(i + 1) % n] - p);
        if (x < 0) ans -= area;
        else ans += area;
    }
    return abs(ans);
}
// find a circle of radius r that contains as many points as
// possible
// 0(n^2 log n);
double maximum_circle_cover(vector<PT> p, double r, circle &
    c) {
    int n = p.size();
    int ans = 0;
    int id = 0; double th = 0;
    for (int i = 0; i < n; ++i) {
        // maximum circle cover when the circle goes through
        // this point
        vector<pair<double, int>> events = {{-PI, +1}, {PI,
            -1}};
        for (int j = 0; j < n; ++j) {
            if (j == i) continue;
            double d = dist(p[i], p[j]);
            if (d > r * 2) continue;
            double dir = (p[j] - p[i]).arg();
            double ang = acos(d / 2 / r);
            double st = dir - ang, ed = dir + ang;
            if (st > PI) st -= PI * 2;
            if (st <= -PI) st += PI * 2;
            if (ed > PI) ed -= PI * 2;
            if (ed <= -PI) ed += PI * 2;

```

```

events.push_back({st - eps, +1}); // take care of
precisions!
events.push_back({ed, -1});
if (st > ed) {
    events.push_back({-PI, +1});
    events.push_back({+PI, -1});
}
}
sort(events.begin(), events.end());
int cnt = 0;
for (auto &&e: events) {
    cnt += e.second;
    if (cnt > ans) {
        ans = cnt;
        id = i; th = e.first;
    }
}
}
PT w = PT[p[id].x + r * cos(th), p[id].y + r * sin(th)];
c = circle(w, r); //best_circle
return ans;
}
// radius of the maximum inscribed circle in a convex
polygon
double maximum_inscribed_circle(vector<PT> p) {
    int n = p.size();
    if (n <= 2) return 0;
    double l = 0, r = 20000;
    while (r - l > eps) {
        double mid = (l + r) * 0.5;
        vector<HP> h;
        const int L = 1e9;
        h.push_back(HP(PT(-L, -L), PT(L, -L)));
        h.push_back(HP(PT(L, -L), PT(L, L)));
        h.push_back(HP(PT(L, L), PT(-L, L)));
        h.push_back(HP(PT(-L, L), PT(-L, -L)));
        for (int i = 0; i < n; i++) {
            PT z = (p[(i + 1) % n] - p[i]).perp();
            z = z.truncate(mid);
            PT y = p[i] + z, q = p[(i + 1) % n] + z;
            h.push_back(HP(p[i] + z, p[(i + 1) % n] + z));
        }
        vector<PT> nw = half_plane_intersection(h);
        if (!nw.empty()) l = mid;
        else r = mid;
    }
    return l;
}

```

6.4 Nearest point

```

const int N = 1<<17;
const long long INF = 2000000000000000000;
struct vp_node {
    long long threshold;
    pair < int, int > center;
    vp_node *left, *right;
};

typedef vp_node *node_ptr;
int tests, n;
pair < int, int > pts[N], arr[N];
long long ans;
node_ptr root;

long long squared_distance(pair < int, int > a, pair < int,
int > b) {
    return (a.first-b.first)*1ll*(a.first-b.first) + (a.
second-b.second)*1ll*(a.second-b.second);
}

struct cmp_points {
    pair < int, int > a;
    cmp_points(){}
    cmp_points(pair < int, int > p): a(p) {}
    bool operator () (const pair < int, int > &x, const pair
< int, int > &y) const {
        return squared_distance(a,x)<squared_distance(a,y);
    }
};

void build(node_ptr &node, int a, int b) {
    if(a>b) {
        node=NULL;
        return;
    }
    node=new vp_node();
    if(a==b) {
        node->threshold=0;
        node->center=arr[a];
        node->left=NULL;
        node->right=NULL;
        return;
    }
    int pos=a+rand()%(b-a+1);
    swap(arr[a],arr[pos]);
    node->center=arr[a];
    sort(arr+a+1,arr+b+1,cmp_points(arr[a]));
    node->threshold=squared_distance(arr[a],arr[(a+b)>>1]);
}

```

```

build(node->left,a,(a+b)>>1);
build(node->right,((a+b)>>1)+1,b);
}

void query(node_ptr &node, pair < int, int > q, long long &
ans) {
    if(node==NULL) return;
    if(node->center!=q) ans=min(ans,squared_distance(node->
center,q));
    if(node->left==NULL && node->right==NULL) {
        return;
    }
    if(squared_distance(q,node->center)<=node->threshold) {
        query(node->left,q,ans);
        if(sqrt(squared_distance(node->center,q))+sqrt(ans)>
sqrt(node->threshold))
            query(node->right,q,ans);
    }
    else {
        query(node->right,q,ans);
        if(sqrt(squared_distance(node->center,q))-sqrt(ans)<
sqrt(node->threshold))
            query(node->left,q,ans);
    }
}
}

```

7 Graph

7.1 articulation pointBridge

```

int n; // number of nodes
vector<vector<int>> adj; // adjacency list of graph
vector<bool> visited;
vector<int> tin, low;
int timer;
void dfs(int v, int p = -1) {
    visited[v] = true;
    tin[v] = low[v] = timer++;
    int children=0;
    for (int to : adj[v]) {
        if (to == p) continue;
        if (visited[to]) {
            low[v] = min(low[v], tin[to]);
        } else {
            dfs(to, v);
            low[v] = min(low[v], low[to]);
            if (low[to] >= tin[v] && p!=-1)
                IS_CUTPOINT(v);
        }
    }
}

```

```

        ++children;
    }
}
if(p == -1 && children > 1)
    IS_CUTPOINT(v);
}
void find_cutpoints() {
    timer = 0;
    visited.assign(n, false);
    tin.assign(n, -1);
    low.assign(n, -1);
    for (int i = 0; i < n; ++i) {
        if (!visited[i])
            dfs(i);
    }
}
// finding bridge.
void IS_BRIDGE(int v, int to); // some function to process
    the found bridge
int n; // number of nodes
vector<vector<int>> adj; // adjacency list of graph
vector<bool> visited;
vector<int> tin, low;
int timer;
void dfs(int v, int p = -1) {
    visited[v] = true;
    tin[v] = low[v] = timer++;
    bool parent_skipped = false;
    for (int to : adj[v]) {
        if (to == p && !parent_skipped) {
            parent_skipped = true;
            continue;
        }
        if (visited[to]) {
            low[v] = min(low[v], tin[to]);
        } else {
            dfs(to, v);
            low[v] = min(low[v], low[to]);
            if (low[to] > tin[v])
                IS_BRIDGE(v, to);
        }
    }
}
void find_bridges() {
    timer = 0;
    visited.assign(n, false);
    tin.assign(n, -1);
    low.assign(n, -1);
    for (int i = 0; i < n; ++i) {
        if (!visited[i])

```

```

        dfs(i);
    }
}

7.2 bell

void solve()
{
    vector<int> d(n, INF);
    d[v] = 0;
    vector<int> p(n, -1);
    int x;
    for (int i = 0; i < n; ++i) {
        x = -1;
        for (Edge e : edges)
            if (d[e.a] < INF)
                if (d[e.b] > d[e.a] + e.cost) {
                    d[e.b] = max(-INF, d[e.a] + e.cost);
                    p[e.b] = e.a;
                    x = e.b;
                }
    }
    if (x == -1)
        cout << "No negative cycle from " << v;
    else {
        int y = x;
        for (int i = 0; i < n; ++i)
            y = p[y];
        vector<int> path;
        for (int cur = y;; cur = p[cur]) {
            path.push_back(cur);
            if (cur == y && path.size() > 1)
                break;
        }
        reverse(path.begin(), path.end());
        cout << "Negative cycle: ";
        for (int u : path)
            cout << u << ' ';
    }
}
}

7.3 kosaraju

```

```

const int N = 1e5 + 5;
vector<int> adj[N], revadj[N], conadj[N];
vector<ll> value(N), concoin(N);
vector<int> component;

```

```

stack<int> st;
vector<bool> vis(N);

void dfs(int u) {
    vis[u] = 1;
    for (auto v : adj[u]) if (!vis[v]) dfs(v);
    st.push(u);
}

void dfs1(int u) {
    vis[u] = 1;
    for (auto v : revadj[u]) if (!vis[v]) dfs1(v);
    component.push_back(u);
}

void dfs2(int u) {
    vis[u] = 1;
    ll mx = 0;
    for (auto v : conadj[u]) {
        if (!vis[v]) dfs2(v);
        mx = max(mx, value[v]);
    }
    value[u] = mx + concoin[u];
}

vector<int> coin(n + 5);
for (int i = 1; i <= n; i++) cin >> coin[i];
for (int i = 1; i <= m; i++) {
    adj[u].push_back(v);
    revadj[v].push_back(u);
}
for (int i = 1; i <= n; i++) if (!vis[i]) dfs(i);
vector<int> roots(n + 5, 0);
vector<int> connodes;
for (int i = 1; i <= n; i++) vis[i] = 0;
while (!st.empty()) {
    int tp = st.top();
    st.pop();
    if (!vis[tp]) {
        dfs1(tp);
        int root = *min_element(component.begin(), component.
            end());
        ll sum = 0;
        for (auto node : component) roots[node] = root, sum
            += coin[node];
        concoin[root] = sum;
        connodes.push_back(root);
        component.clear();
    }
}
for (int u = 1; u <= n; u++) {

```



```

    for (auto v : adj[u]) {
        if (roots[u] != roots[v]) conadj[roots[u]].push_back(
            roots[v]);
    }
}
for (int i = 1; i <= n; i++) vis[i] = 0;
for (auto u : connodes) if (!vis[u]) dfs2(u);

```

8 Number Theory Math

8.1 CRT

```

__int128 ExU(__int128 a, __int128 b, __int128 &x, __int128 &y) {
    if (b == 0) {
        x = 1;
        y = 0;
        return a;
    }
    __int128 x1, y1;
    __int128 d = ExU(b, a % b, x1, y1);
    x = y1;
    y = x1 - (a / b) * y1;
    return d;
}
void solve() {
    cout << "Case " << ii++ << ": ";
    int n; cin >> n;
    vector<int> a(n), m(n);
    for (int i = 0; i < n; i++) cin >> m[i] >> a[i];
    int a1 = a[0], m1 = m[0];
    for (int i = 1; i < n; i++) {
        __int128 a2 = a[i], m2 = m[i];
        __int128 x, y;
        __int128 g = ExU(m1, m2, x, y);
        if ((a1 - a2) % g != 0) {
            cout << "Impossible" << endl;
            return;
        }
        __int128 k = (a1 - a2) / g, lcm = (m1 / g) * m2;
        __int128 a = a1 - m1 * (k * x);
        a1 = (a % lcm);
        if (a1 < 0) a1 += lcm;
        m1 = lcm;
    }
    cout << a1 << endl;
}

```

8.2 Diophantile

```

int gcd(int a, int b, int& x, int& y) {
    if (b == 0) { x = 1; y = 0; return a; }
}
int x1, y1;
int d = gcd(b, a % b, x1, y1);
x = y1; y = x1 - y1 * (a / b);
return d;
}
bool find_any_solution(int a, int b, int c, int &x0, int &y0,
    int &g) {
    g = gcd(abs(a), abs(b), x0, y0);
    if (c % g) {
        return false;
    }
    x0 *= c / g; y0 *= c / g;
    if (a < 0) x0 = -x0;
    if (b < 0) y0 = -y0;
    return true;
}
void shift_solution(int &x, int &y, int a, int b, int cnt) {
    x += cnt * b; y -= cnt * a;
}
int find_all_solutions(int a, int b, int c, int minx, int maxx, int miny, int maxy) {
    int x, y, g;
    if (!find_any_solution(a, b, c, x, y, g)) return 0;
    a /= g; b /= g;
    int sign_a = a > 0 ? +1 : -1;
    int sign_b = b > 0 ? +1 : -1;
    shift_solution(x, y, a, b, (minx - x) / b);
    if (x < minx) shift_solution(x, y, a, b, sign_b);
    if (x > maxx) return 0;
    int lx1 = x;
    shift_solution(x, y, a, b, (maxx - x) / b);
    if (x > maxx) shift_solution(x, y, a, b, -sign_b);
    int rx1 = x; shift_solution(x, y, a, b, -(miny - y) / a);
    if (y < miny) shift_solution(x, y, a, b, -sign_a);
    if (y > maxy) return 0;
    int lx2 = x;
    shift_solution(x, y, a, b, -(maxy - y) / a);
    if (y > maxy) shift_solution(x, y, a, b, sign_a);
    int rx2 = x;
    if (lx2 > rx2) swap(lx2, rx2);
    int lx = max(lx1, lx2);
    int rx = min(rx1, rx2);
    if (lx > rx) return 0;
    return (rx - lx) / abs(b) + 1;
}

```

```

}

```

8.3 FFT

```

const int N = 3e5 + 9;

const double PI = acos(-1);
struct base {
    double a, b;
    base(double a = 0, double b = 0) : a(a), b(b) {}
    const base operator + (const base &c) const {
        { return base(a + c.a, b + c.b); }
    }
    const base operator - (const base &c) const {
        { return base(a - c.a, b - c.b); }
    }
    const base operator * (const base &c) const {
        { return base(a * c.a - b * c.b, a * c.b + b * c.a); }
    }
};
void fft(vector<base> &p, bool inv = 0) {
    int n = p.size(), i = 0;
    for (int j = 1; j < n - 1; ++j) {
        for (int k = n >> 1; k > (j ^ k); k >>= 1);
        if (j < i) swap(p[i], p[j]);
    }
    for (int l = 1, m; (m = l << 1) <= n; l <= m) {
        double ang = 2 * PI / m;
        base wn = base(cos(ang), (inv ? 1. : -1.) * sin(ang)), w;
        for (int i = 0, j, k; i < n; i += m) {
            for (w = base(1, 0), j = i, k = i + 1; j < k; ++j, w = w * wn) {
                base t = w * p[j + 1];
                p[j + 1] = p[j] - t;
                p[j] = p[j] + t;
            }
        }
    }
    if (inv) for (int i = 0; i < n; ++i) p[i].a /= n, p[i].b /= n;
}
vector<long long> multiply(vector<int> &a, vector<int> &b) {
    int n = a.size(), m = b.size(), t = n + m - 1, sz = 1;
    while (sz < t) sz <= 1;
    vector<base> x(sz), y(sz), z(sz);
    for (int i = 0; i < sz; ++i) {
        x[i] = i < (int)a.size() ? base(a[i], 0) : base(0, 0);
        y[i] = i < (int)b.size() ? base(b[i], 0) : base(0, 0);
    }
    fft(x), fft(y);
    for (int i = 0; i < sz; ++i) z[i] = x[i] * y[i];
    fft(z, 1);
}

```



```

vector<long long> ret(sz);
for(int i = 0; i < sz; ++i) ret[i] = (long long) round(z[i]
    ].a);
while((int)ret.size() > 1 && ret.back() == 0) ret.pop_back
    ();
return ret;
}

```

8.4 LSieveBigModTernaryS

```

int a[], n elements;
int mx = 0; int ans = 0;
for (int i = 0; i < n; i++) {
    mx = max(mx+a[i], (int)0);
    ans = max(ans, mx);
}
#define sz 1000
vector<int> prime;
bitset<sz> bs;
void sieve() {
    bs.set(); // all bit = 1..
    bs[0] = bs[1] = 0;
    prime.push_back(2);
    for (int i = 3; i <= sz; i += 2) {
        if(bs[i]) {
            for (int j = i*i; j <= sz; j += i)
                bs[j] = 0;
            prime.push_back(i);
        }
    }
}
void solve()
{
    int n = 12;
    int a[] = {60, 70, 80, 90, 100, 200, 400, 500, 600, 700,
        900, 1000};
    int r = n-1; int l = 0;
    // cout << (r-l) / 2;
    while(l < r) {
        int c = (r-l) / 3;
        int m1 = l + c;
        int m2 = r - c;
        // cout << a[m1] << ' ' << a[m2] << endl;
        if(a[m1] < a[m2]) {
            r = m2 - 1;
        } else if(a[m1] > a[m2]) {
            l = m1 + 1;
        } else {
            l = m1 + 1; r = m2 - 1;
        }
    }
}

```

```

    }
}
cout << a[l] << endl;
}
int mod = 1e9 + 7;
// linear sieve.
vector<int> prime;
bool is_composite[MAXN];
void sieve (int n) {
    std::fill (is_composite, is_composite + n, false);
    for (int i = 2; i < n; ++i) {
        if (!is_composite[i]) prime.push_back (i);
        for (int j = 0; j < prime.size () && i * prime[j] < n; ++j)
            {
                is_composite[i * prime[j]] = true;
                if (i % prime[j] == 0) break;
            }
    }
}
}

```

8.5 matexpo

```

const int M = 1e9 + 7;
vector<vector<int>> multiply(vector<vector<int>> a, vector<
    vector<int>> b) {
    int sz = a.size();
    vector<vector<int>> c(sz, vector<int> (sz, 0));
    for (int i = 0; i < sz; i++)
        for (int j = 0; j < sz; j++)
            for (int k = 0; k < sz; k++)
                c[i][j] = (c[i][j] + 1ll * a[i][k] * b[k][j])
                    % M;

    return c;
}
vector<vector<int>> binexpo(vector<vector<int>> m, ll n) {
    vector<vector<int>> res
        = {{1, 0, 0, 0}, {0, 1, 0, 0}, {0, 0, 1, 0}, {0, 0, 0,
            1}};

    while(n > 0) {
        if(n & 1ll) res = multiply(res, m);
        m = multiply(m, m);
        n >>= 1;
    }
    return res;
}
vector<vector<int>> m
    = {{3, 2, 1, 3}, {1, 0, 0, 0}, {0, 1, 0, 0}, {0, 0, 0, 1}};

```

```

vector<vector<int>> b
    = {{3, 0, 0, 0}, {2, 0, 0, 0}, {1, 0, 0, 0}, {1, 0, 0, 0}
    };
auto res = binexpo(m, k - 2);
res = multiply(res, b);

```

8.6 Miller Rabin

```

using u64 = uint64_t;
using u128 = __uint128_t;
u64 binpower(u64 base, u64 e, u64 mod) {
    u64 result = 1;
    base %= mod;
    while (e) {
        if (e & 1)
            result = (u128)result * base % mod;
        base = (u128)base * base % mod;
        e >>= 1;
    }
    return result;
}
bool check_composite(u64 n, u64 a, u64 d, int s) {
    u64 x = binpower(a, d, n);
    if (x == 1 || x == n - 1)
        return false;
    for (int r = 1; r < s; r++) {
        x = (u128)x * x % n;
        if (x == n - 1)
            return false;
    }
    return true;
};
bool MillerRabin(u64 n, int iter=5) { // returns true if n
    is probably prime, else returns false.
    if (n < 4)
        return n == 2 || n == 3;
    int s = 0;
    u64 d = n - 1;
    while ((d & 1) == 0) {
        d >>= 1;
        s++;
    }
    for (int i = 0; i < iter; i++) {
        int a = 2 + rand() % (n - 3);
        if (check_composite(n, a, d, s))
            return false;
    }
    return true;
}

```

```
bool MillerRabin(u64 n) { // returns true if n is prime,
    else returns false.
    if (n < 2)
        return false;
    int r = 0;
    u64 d = n - 1;
    while ((d & 1) == 0) {
        d >>= 1;
        r++;
    }
    for (int a : {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31,
        37}) {
        if (n == a)
            return true;
        if (check_composite(n, a, d, r))
            return false;
    }
    return true;
}
```

8.7 mobius

```
int mob[N];
void mobius() {
    mob[1] = 1;
    for (int i = 2; i < N; i++){
        mob[i]--;
        for (int j = i + i; j < N; j += i) {
            mob[j] -= mob[i];
        }
    }
}
bool vis[N];
vector<int> d[N];
int mul[N];
void add(int x, int k) {
    for (auto y: d[x]) {
        mul[y] += k;
    }
}
int query(int x) {
    int ans = 0;
    for (auto y: d[x]) {
        ans += mul[y] * mob[y];
    }
    return ans;
}
```

8.8 ncr

```
//nCr: (n,0)=1,(n,1)=n,(n,r)=(n-1,r)+(n-1,r-1)
//Circular permutation: nPr/r
ll fact[MAX], inv[MAX], invFact[MAX];
ll nCr(ll n, ll r) {
    if (r > n) return 0;
    return fact[n] * invFact[r] % MOD * invFact[n - r] % MOD;
}
void PreCalcalaton() {
    fact[0] = 1;
    for (i = 1; i < MAX; ++i) fact[i] = i * fact[i - 1] % MOD;
    inv[1] = 1;
    for (i = 2; i < MAX; ++i) //must be MAX<MOD
        inv[i] = ((MOD - MOD / i) * inv[MOD % i]) % MOD;
    invFact[0] = 1;
    for (i = 1; i < MAX; ++i)
        invFact[i] = (invFact[i - 1] * inv[i]) % MOD;
}
//nCr_mod_p_by_lucas_for_small_p._0(LogN)
ll Lucas(ll n, ll m) {
    if (n == 0 && m == 0) return 1ll;
    ll ni = n % MOD; ll mi = m % MOD;
    if (mi > ni) return 0;
    return (Lucas(n / MOD, m / MOD) * nCr(ni, mi)) % MOD;
}
ll nCr_by_lucas(ll n, ll r) { return Lucas(n, r); }
//nCr_mod_M, M_is_not_prime
const int N = 142858; //need primes <=M (nCr%M)
int spf[N]; vector<int> primes;
void sieve() {
    for (int i = 2; i < N; i++) {
        if (spf[i] == 0) spf[i] = i, primes.push_back(i);
        int sz = primes.size();
        for (int j = 0; j < sz && i * primes[j] < N && primes[j]
            <= spf[i]; j++)
            spf[i * primes[j]] = primes[j];
    }
}
// returns n! % mod, mod = multiple of p
// 0(mod * log(n))
int factmod(ll n, int p, const int mod) {
    vector<int> f(mod + 1); f[0] = 1 % mod;
    for (int i = 1; i <= mod; i++) {
        if (i % p) f[i] = 1LL * f[i - 1] * i % mod;
        else f[i] = f[i - 1];
    }
    int ans = 1 % mod;
    while (n > 1) {
        ans = 1LL * ans * f[n % mod] % mod;

```

```
ans = 1LL * ans * BigMod(f[mod], n / mod, mod) % mod;
n /= p;
    } return ans;
}
ll multiplicity(ll n, int p) {
    ll ans = 0;
    while (n) { n /= p; ans += n; } return ans;
}
// C(n, r) modulo p^k
// 0(p^k log n)
int ncr(ll n, ll r, int p, int k) {
    if (n < r || r < 0) return 0;
    int mod = 1;
    for (int i = 0; i < k; i++) { mod *= p; }
    ll t = multiplicity(n, p) - multiplicity(r, p) -
        multiplicity(n - r, p);
    if (t >= k) return 0;
    int ans = 1LL * factmod(n, p, mod) * inverse(factmod(r, p,
        mod), mod) % mod * inverse(factmod(n - r, p, mod),
        mod) % mod;
    ans = 1LL * ans * BigMod(p, t, mod) % mod;
    return ans;
}
pair<ll, ll> CRT(ll a1, ll m1, ll a2, ll m2) {
    ll p, q;
    ll g = e_gcd(m1, m2, p, q);
    if (a1 % g != a2 % g) return make_pair(0, -1);
    ll m = m1 / g * m2;
    p = (p % m + m) % m; q = (q % m + m) % m;
    return make_pair((p * a2 % m * (m1 / g) % m + q * a1 % m *
        (m2 / g) % m) % m, m);
}
// 0(m log(n) log(m))
int ncr(ll n, ll r, int m) {
    if (n < r || r < 0) return 0;
    pair<ll, ll> ans({0, 1});
    while (m > 1) {
        int p = spf[m], k = 0, cur = 1;
        while (m % p == 0) {
            m /= p; cur *= p; ++k;
        }
        ans = CRT(ans.first, ans.second, ncr(n, r, p, k), cur);
    } return ans.first;
}
```

8.9 totient

```
int phi(int n) {
    int result = n;

```

```

    for (int i = 2; i * i <= n; i++) {
        if (n % i == 0) {
            while (n % i == 0)
                n /= i;
            result -= result / i;
        }
    }
    if (n > 1)
        result -= result / n;
    return result;
}

void phi_1_to_n(int n) {
    vector<int> phi(n + 1);
    for (int i = 0; i <= n; i++)
        phi[i] = i;

    for (int i = 2; i <= n; i++) {
        if (phi[i] == i) {
            for (int j = i; j <= n; j += i)
                phi[j] -= phi[j] / i;
        }
    }
}

void phi_1_to_n(int n) {
    vector<int> phi(n + 1);
    phi[0] = 0;
    phi[1] = 1;
    for (int i = 2; i <= n; i++)
        phi[i] = i - 1;

    for (int i = 2; i <= n; i++)
        for (int j = 2 * i; j <= n; j += i)
            phi[j] -= phi[i];
}

```

9 String

9.1 AhoCorasick

```

typedef struct AC {
    static const int K = 26;
    struct node {
        int nxt[K], link, leaf, par;
        vector<int> ind;
        char pch;
        node(int par = -1, char pch = '$') : par(par), pch(
            pch) {
            fill(begin(nxt), end(nxt), -1);

```

```

        leaf = false;
        link = -1;
    }
};

vector<node> t;
vector<vector<int>> ad;
vector<int> st, en, mp;
vector<int> cnt;
int dt = 0;
AC() {
    mp = vector<int> (1<<8, 0);
    for (char ch = 'a'; ch <= 'z'; ++ch) mp[ch] = ch - 'a';
    t.resize(1);
    cnt.resize(1);
}

int add(string str, int i) {
    int ptr = 0;
    for (auto ch : str) {
        if (t[ptr].nxt[mp[ch]] < 0) {
            t[ptr].nxt[mp[ch]] = t.size();
            t.push_back(node(ptr, ch));
            cnt.push_back(0);
        }
        ptr = t[ptr].nxt[mp[ch]];
    }
    t[ptr].leaf = true;
    t[ptr].ind.push_back(i);
    return ptr;
}

int get_link(int v) {
    if (t[v].link == -1) {
        t[v].link = (!v || !t[v].par) ? 0 : go(get_link(t[v].par), t[v].pch);
    }
    return t[v].link;
}

int go(int v, char ch) {
    if (t[v].nxt[mp[ch]] < 0) {
        t[v].nxt[mp[ch]] = !v ? 0 : go(get_link(v), ch);
    }
    return t[v].nxt[mp[ch]];
}

void dfs(int v) {
    st[v] = ++dt;
    for (int u : ad[v]) {
        dfs(u);
        cnt[v] += cnt[u];
    }
    en[v] = dt;

```

```

}

void calc() {
    int k = t.size();
    st.resize(k);
    en.resize(k);
    ad.resize(k);
    for (int i = 1; i < k; i++)
        ad[get_link(i)].push_back(i);
    dfs(0);
}

} cc;
int cs = 1;
void solve() {
    cc bc;
    int n; cin >> n;
    vector<int> res(n + 1, 0);
    string text;
    cin >> text;
    string p;
    for (int i = 1; i <= n; i++) {
        cin >> p;
        bc.add(p, i);
    }
    int vertex = 0;
    for (int i = 0; i < text.length(); i++) {
        vertex = bc.go(vertex, text[i]);
        bc.cnt[vertex]++;
    }
    bc.calc();
    for (int i = 1; i < bc.t.size(); i++) {
        for (int &x : bc.t[i].ind) res[x] += bc.cnt[i];
    }
    cout << "Case " << cs++ << ":\n";
    for (int i = 1; i <= n; i++) cout << res[i] << "\n";
}

```

9.2 hashing

```

const int N = 2e5 + 10;
vector<int> p1(N, 1), p2(N, 1);
vector<int> inp1(N, 1), inp2(N, 1);
int b1 = 37, m1 = 1e9 + 7;
int b2 = 73, m2 = 998244341;
int modp(int i, int j, int m) {
    return 1ll * i * j % m;
}
int moda(int i, int j, int m) {
    return (i + j) % m;
}

```

```

int mods(int i, int j, int m) {
    return ((i - j) % m + m) % m;
}

vector<int> get_hash(string s, int b, int m, vector<int> &p)
{
    //hash vector of a string 1000456781 998244341 37 73
    int sz = s.size();
    s = '#' + s; //everything is 1 based indexing
    vector<int> hash(sz + 10);
    hash[0] = 0;
    hash[1] = (s[1] - 'a' + 1);
    for (int i = 1; i <= sz; i++) {
        int cur = modp(s[i] - 'a' + 1, p[i], m);
        hash[i] = moda(hash[i - 1], cur, m) % m;
    }
    return hash;
}

int getHashL_R(int l, int r, vector<int> &hash, vector<int>
    &inp, int m) {
    int hsh = mods(hash[r], hash[l - 1], m);
    hsh = modp(hsh, inp[l], m);
    return hsh;
}

int binpow(int a, int b, int m) {
    a %= m;
    int res = 1;
    while (b > 0) {
        if (b & 1) res = 1ll * res * a % m;
        a = 1ll * a * a % m;
        b >>= 1;
    }
    return res % m;
}

void pre() {
    //p[i] have power i - 1;
    for (int i = 2; i < N; i++) {
        p1[i] = modp(p1[i - 1], b1, m1);
        p2[i] = modp(p2[i - 1], b2, m2);
        inp1[i] = binpow(p1[i], m1 - 2, m1);
        inp2[i] = binpow(p2[i], m2 - 2, m2);
    }
}

```

9.3 Manacher

```

struct Manacher {
    vector<int> p[2];
    // p[1][i] = (max odd length palindrome centered at i) / 2
    [floor division]

```

```

// p[0][i] = same for even, it considers the right center
// e.g. for s = "abbabba", p[1][3] = 3, p[0][2] = 2
Manacher(string s) {
    int n = s.size();
    p[0].resize(n + 1);
    p[1].resize(n);
    for (int z = 0; z < 2; z++) {
        for (int i = 0, l = 0, r = 0; i < n; i++) {
            int t = r - i + !z;
            if (i < r) p[z][i] = min(t, p[z][l + t]);
            int L = i - p[z][i], R = i + p[z][i] - !z;
            while (L >= 1 && R + 1 < n && s[L - 1] == s[R + 1])
                p[z][i]++, L--, R++;
            if (R > r) l = L, r = R;
        }
    }
}

bool is_palindrome(int l, int r) {
    int mid = (l + r + 1) / 2, len = r - l + 1;
    return 2 * p[len % 2][mid] + len % 2 >= len;
}

int32_t main() {
    string s; cin >> s;
    Manacher M(s);
    int n = s.size();
    for (int i = 0; i < n; i++) {
        cout << 2 * M.p[1][i] + 1 << ' ';
        if (i + 1 < n) cout << 2 * M.p[0][i + 1] << ' ';
    }
    cout << '\n';
    return 0;
}

```

9.4 suffixArray

```

const int N = 3e5 + 9;
const int LG = 18;
void induced_sort(const vector<int> &vec, int val_range,
    vector<int> &SA, const vector<bool> &sl, const vector<
    int> &lms_idx) {
    vector<int> l(val_range, 0), r(val_range, 0);
    for (int c : vec) {
        if (c + 1 < val_range) ++l[c + 1];
        ++r[c];
    }
    partial_sum(l.begin(), l.end(), l.begin());
    partial_sum(r.begin(), r.end(), r.begin());
    fill(SA.begin(), SA.end(), -1);

```

```

for (int i = lms_idx.size() - 1; i >= 0; --i)
    SA[--r[vec[lms_idx[i]]]] = lms_idx[i];
for (int i : SA)
    if (i >= 1 && sl[i - 1]) {
        SA[l[vec[i - 1]]++] = i - 1;
    }
fill(r.begin(), r.end(), 0);
for (int c : vec)
    ++r[c];
partial_sum(r.begin(), r.end(), r.begin());
for (int k = SA.size() - 1, i = SA[k]; k >= 1; --k, i = SA
    [k])
    if (i >= 1 && !sl[i - 1]) {
        SA[--r[vec[i - 1]]] = i - 1;
    }
}

vector<int> SA_IS(const vector<int> &vec, int val_range) {
    const int n = vec.size();
    vector<int> SA(n), lms_idx;
    vector<bool> sl(n);
    sl[n - 1] = false;
    for (int i = n - 2; i >= 0; --i) {
        sl[i] = (vec[i] > vec[i + 1] || (vec[i] == vec[i + 1] &&
            sl[i + 1]));
        if (sl[i] && !sl[i + 1]) lms_idx.push_back(i + 1);
    }
    reverse(lms_idx.begin(), lms_idx.end());
    induced_sort(vec, val_range, SA, sl, lms_idx);
    vector<int> new_lms_idx(lms_idx.size(), lms_vec[lms_idx.
        size()]);
    for (int i = 0, k = 0; i < n; ++i)
        if (!sl[SA[i]] && SA[i] >= 1 && sl[SA[i] - 1]) {
            new_lms_idx[k++] = SA[i];
        }
    int cur = 0;
    SA[n - 1] = cur;
    for (size_t k = 1; k < new_lms_idx.size(); ++k) {
        int i = new_lms_idx[k - 1], j = new_lms_idx[k];
        if (vec[i] != vec[j]) {
            SA[j] = ++cur;
            continue;
        }
        bool flag = false;
        for (int a = i + 1, b = j + 1; ++a, ++b) {
            if (vec[a] != vec[b]) {
                flag = true;
                break;
            }
        }
        if ((!sl[a] && sl[a - 1]) || (!sl[b] && sl[b - 1])) {

```

```

        flag = !((!sl[a] && sl[a - 1]) && (!sl[b] && sl[b - 1]));
        break;
    }
    SA[j] = (flag ? ++cur : cur);
}
for (size_t i = 0; i < lms_idx.size(); ++i)
    lms_vec[i] = SA[lms_idx[i]];
if (cur + 1 < (int)lms_idx.size()) {
    auto lms_SA = SA_IS(lms_vec, cur + 1);
    for (size_t i = 0; i < lms_idx.size(); ++i) {
        new_lms_idx[i] = lms_idx[lms_SA[i]];
    }
}
induced_sort(vec, val_range, SA, sl, new_lms_idx);
return SA;
}
vector<int> suffix_array(const string &s, const int LIM =
    128) {
    vector<int> vec(s.size() + 1);
    copy(begin(s), end(s), begin(vec));
    vec.back() = '1';
    auto ret = SA_IS(vec, LIM);
    ret.erase(ret.begin());
    return ret;
}
struct SuffixArray {
    int n;
    string s;
    vector<int> sa, rank, lcp;
    vector<vector<int>> t;
    vector<int> lg;
    SuffixArray() {}
    SuffixArray(string _s) {
        n = _s.size();
        s = _s;
        sa = suffix_array(s);
        rank.resize(n);
        for (int i = 0; i < n; i++) rank[sa[i]] = i;
        construct_lcp();
        prec();
        build();
    }
    void construct_lcp() {
        int k = 0;
        lcp.resize(n - 1, 0);
        for (int i = 0; i < n; i++) {
            if (rank[i] == n - 1) {
                k = 0;

```

```

                continue;
            }
            int j = sa[rank[i] + 1];
            while (i + k < n && j + k < n && s[i + k] == s[j + k])
                k++;
            lcp[rank[i]] = k;
            if (k) k--;
        }
    }
    void prec() {
        lg.resize(n, 0);
        for (int i = 2; i < n; i++) lg[i] = lg[i / 2] + 1;
    }
    void build() {
        int sz = n - 1;
        t.resize(sz);
        for (int i = 0; i < sz; i++) {
            t[i].resize(LG);
            t[i][0] = lcp[i];
        }
        for (int k = 1; k < LG; ++k) {
            for (int i = 0; i + (1 << k) - 1 < sz; ++i) {
                t[i][k] = min(t[i][k - 1], t[i + (1 << (k - 1))][k - 1]);
            }
        }
    }
    int query(int l, int r) { // minimum of lcp[l], ..., lcp[r]
        int k = lg[r - l + 1];
        return min(t[l][k], t[r - (1 << k) + 1][k]);
    }
    int get_lcp(int i, int j) { // lcp of suffix starting from
        i and j
        if (i == j) return n - i;
        int l = rank[i], r = rank[j];
        if (l > r) swap(l, r);
        return query(l, r - 1);
    }
    int lower_bound(string &t) {
        int l = 0, r = n - 1, k = t.size(), ans = n;
        while (l <= r) {
            int mid = l + r >> 1;
            if (s.substr(sa[mid], min(n - sa[mid], k)) >= t) ans =
                mid, r = mid - 1;
            else l = mid + 1;
        }
        return ans;
    }
    int upper_bound(string &t) {

```

```

        int l = 0, r = n - 1, k = t.size(), ans = n;
        while (l <= r) {
            int mid = l + r >> 1;
            if (s.substr(sa[mid], min(n - sa[mid], k)) > t) ans =
                mid, r = mid - 1;
            else l = mid + 1;
        }
        return ans;
    }
    // occurrences of s[p, ..., p + len - 1]
    pair<int, int> find_occurrence(int p, int len) {
        p = rank[p];
        pair<int, int> ans = {p, p};
        int l = 0, r = p - 1;
        while (l <= r) {
            int mid = l + r >> 1;
            if (query(mid, p - 1) >= len) ans.first = mid, r = mid
                - 1;
            else l = mid + 1;
        }
        l = p + 1, r = n - 1;
        while (l <= r) {
            int mid = l + r >> 1;
            if (query(p, mid - 1) >= len) ans.second = mid, l = mid
                + 1;
            else r = mid - 1;
        }
        return ans;
    }
};

```

9.5 z kmp

```

vector<int> prefix_function(string s) {
    int n = (int)s.length();
    vector<int> pi(n);
    for (int i = 1; i < n; i++) {
        int j = pi[i - 1];
        while (j > 0 && s[i] != s[j])
            j = pi[j - 1];
        if (s[i] == s[j])
            j++;
        pi[i] = j;
    }
    return pi;
}
vector<int> z_function(string pattern, string text) {
    string s = pattern + "$" + text;
    int n = size(s);

```

```

vector<int> z(n);
for (int i = 1, l = 0, r = 0; i < n; ++i) {
    if (i <= r)
        z[i] = min (r - i + 1, z[i - 1]);
    while (i + z[i] < n && s[z[i]] == s[i + z[i]])
        ++z[i];
    if (i + z[i] - 1 > r)
        l = i, r = i + z[i] - 1;
}
return z;
}

```

10 template

```

#include <bits/stdc++.h>
#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>
#define F first
#define S second
using namespace std;
using namespace __gnu_pbds;
using ll = long long;
using pii = array<int, 2>;
using tup = array<int, 3>;
using LL = __int128;
template <typename T>
using order_set = tree<T, null_type, less_equal<T>,
    rb_tree_tag, tree_order_statistics_node_update>;
template <typename T> using minheap = priority_queue<T,
    vector<T>, greater<T>>;

```

```

//find_by_order() -> iterator to the k-th largest element (
    zero based)
//order_of_key() -> the number of items that are strictly
    smaller
mt19937_64 rng(chrono::steady_clock::now().time_since_epoch
    ().count());
void solve(int tc) {
}
int32_t main() {
    ios_base::sync_with_stdio(false); cin.tie(0);
    int t = 1, tc = 0; cin >> t;
    while (t--) solve(++tc);
}

```

11 Tree

11.1 eulerTour

```

const int N = 2e5 + 5;
vector<int> adj[N], flat(N), a(N), st(N), en(N);
int t;
void dfs(int u, int p) {
    st[u] = ++t;
    for (auto v : adj[u]) if (v != p) dfs(v, u);
    en[u] = t;
}
dfs(1, 0);
for (int i = 1; i <= n; i++) flat[st[i]] = a[i];
SGTree sg(t);
sg.build(1, 1, t, flat);

```

11.2 lca

```

const int B = 25;
const int N = 2e5 + 5;
vector<int> adj[N];
vector<int> dep(N, -1);
vector<vector<int>> par(B, vector<int> (N, 0));
void dfs(int u, int p) {
    par[0][u] = p;
    dep[u] = dep[p] + 1;
    for (auto v : adj[u]) if (v != p) {
        dfs(v, u);
    }
}
int lca(int u, int v) {
    if (dep[v] > dep[u]) swap(u, v);
    int d = dep[u] - dep[v];
    for (int i = B - 1; i >= 0; --i) {
        if (d & (1 << i)) u = par[i][u];
    }
    for (int i = B - 1; i >= 0; --i) {
        if (par[i][u] != par[i][v]) {
            u = par[i][u];
            v = par[i][v];
        }
    }
    return (u == v ? u : par[0][u]);
}

```